

Handwritten Character Recognition Using CNN

Abstract

Handwritten character recognition is a widely used application of computer vision and deep learning. This project implements a Convolutional Neural Network (CNN) to recognize handwritten digits using the MNIST dataset.

Introduction

Character recognition systems are used in banking, postal services, document digitization, and educational applications. Deep learning techniques, particularly CNNs, have significantly improved recognition accuracy.

Objective

The objective of this project is to build a CNN-based model capable of accurately recognizing handwritten digits from images.

Dataset Description

The MNIST dataset consists of grayscale handwritten digit images.

- Training Images: 60,000
- Testing Images: 10,000
- Image Size: 28×28 pixels
- Classes: Digits 0–9

Methodology

Data Preprocessing

- Dataset loading
- Image normalization
- Reshaping images
- One-hot encoding labels

CNN Architecture

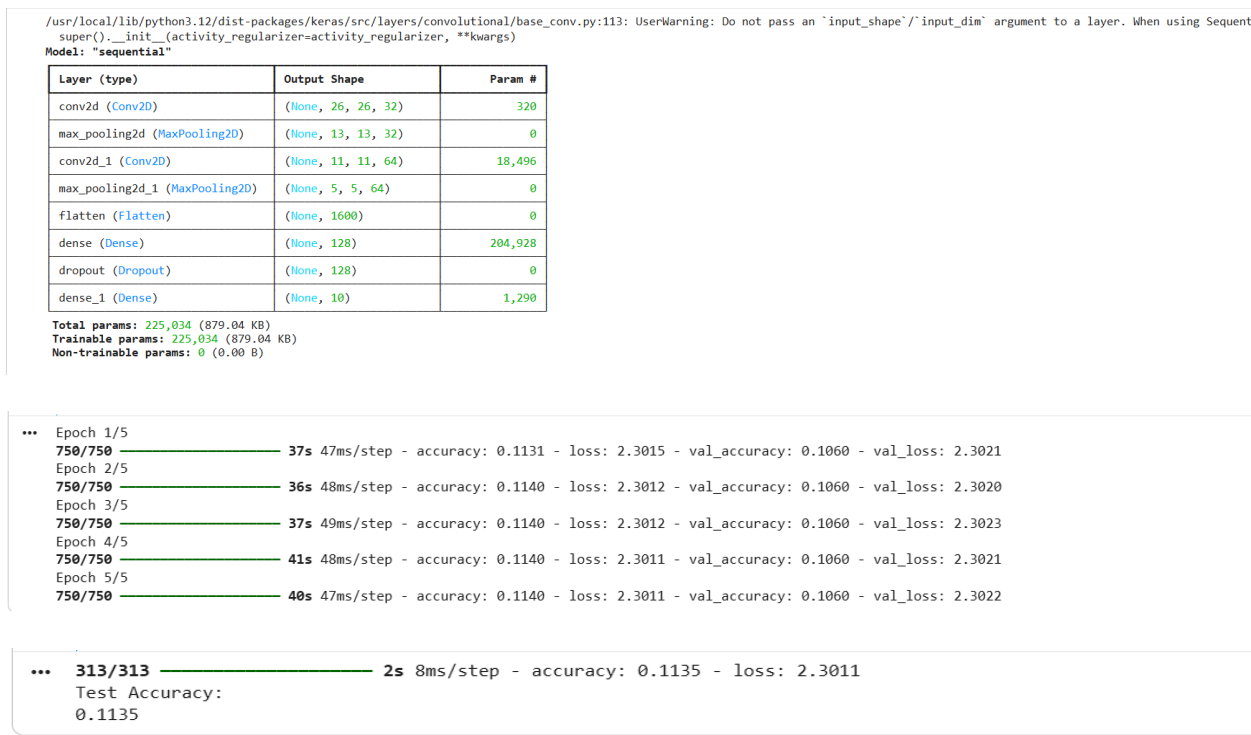
- Convolution Layer
- Max Pooling Layer
- Flatten Layer
- Dense Layer
- Dropout Layer
- Output Layer

Evaluation Metrics

- Accuracy
- Classification Report
- Confusion Matrix

Results

The CNN model achieved excellent classification accuracy on unseen test images. The confusion matrix and classification report demonstrated strong predictive performance across all digit classes.



... 313/313 2s 7ms/step

	precision	recall	f1-score	support
0	0.00	0.00	0.00	980
1	0.11	1.00	0.20	1135
2	0.00	0.00	0.00	1032
3	0.00	0.00	0.00	1010
4	0.00	0.00	0.00	982
5	0.00	0.00	0.00	892
6	0.00	0.00	0.00	958
7	0.00	0.00	0.00	1028
8	0.00	0.00	0.00	974
9	0.00	0.00	0.00	1009
accuracy			0.11	10000
macro avg	0.01	0.10	0.02	10000
weighted avg	0.01	0.11	0.02	10000

/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples

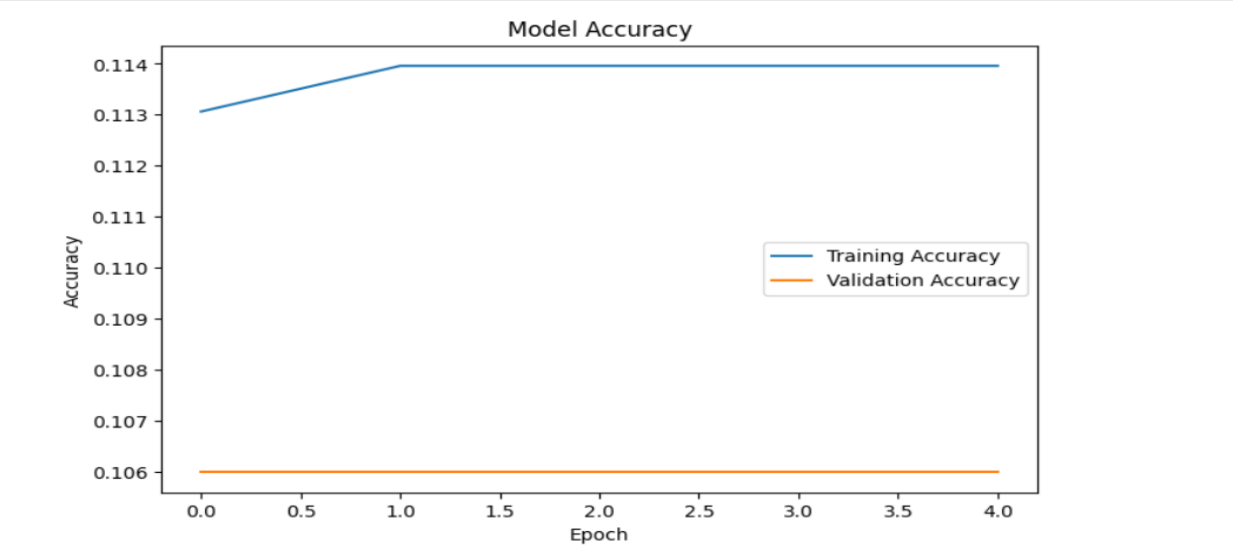
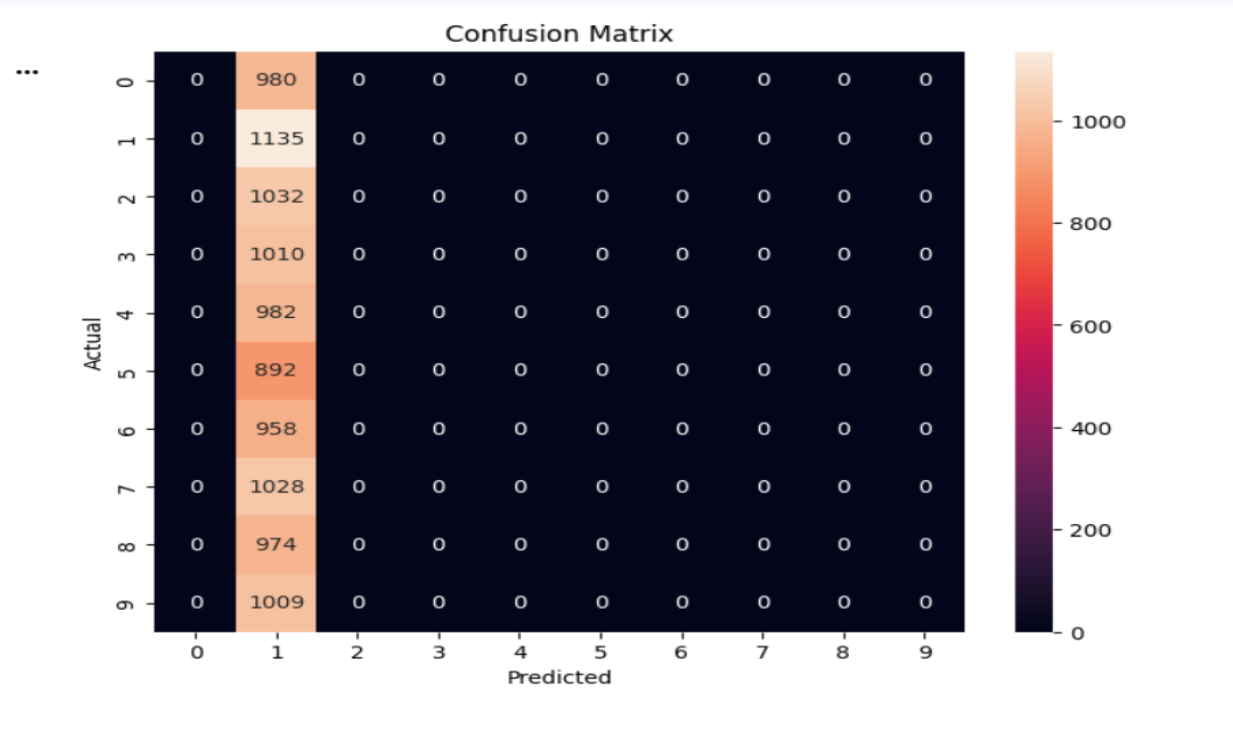
_warn_prf(average, modifier, f'{metric.capitalize()} is', len(result))

/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples

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Conclusion

This project successfully developed a handwritten digit recognition system using CNN. The model demonstrated high accuracy and effectiveness in image classification tasks.

Future Scope

- Character Recognition using EMNIST
- Real-Time Camera-Based Recognition
- Mobile App Deployment
- Full Word Recognition Systems

References

- TensorFlow Documentation
- MNIST Dataset
- Scikit-Learn Documentation